Computation of Light-Cone Superstring States and Factorization of the Universal Massive Supersymmetry Multiplet

Massive Modes Analysis Framework

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Abstract

This document provides the detailed mathematical and physical derivation of the light-cone Green-Schwarz open superstring Fock space, the exact polynomial character factorization of the universal massive supersymmetry factor $\mathcal{M}_{\text{susy}} = 128_B \oplus 128_F$, and the subsequent decomposition into SO(8), SO(9), and $D = 11$ SO(10) supermultiplets as proposed by I. Bars in hep-th/9503228.

1 Light-Cone Green-Schwarz Open Superstring Oscillators

In the light-cone gauge formulation of the 10-dimensional Green-Schwarz (GS) open superstring, physical states are built from transverse bosonic and fermionic harmonic oscillators transforming under the transverse SO(8) rotation group:

1. **Bosonic Oscillators:** α^i_{-n} with mode number $n \geq 1$ and vector index $i = 1, \ldots, 8$, transforming in the vector representation $\mathbf{8}_v$ of SO(8).
2. **Fermionic Oscillators:** S^a_{-n} with mode number $n \geq 1$ and spinor index $a = 1, \ldots, 8$, transforming in the chiral spinor representation $\mathbf{8}_s$ of SO(8).

In the standard orthogonal basis of SO(8) weights (h_1, h_2, h_3, h_4), the weights of the basic representations are given by:

$$\mathbf{8}_v = \{(\pm 1, 0, 0, 0), (0, \pm 1, 0, 0), (0, 0, \pm 1, 0), (0, 0, 0, \pm 1)\}, \tag{1}$$

$$\mathbf{8}_s = \left\{ \left(\pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2}\right) \;\middle|\; \prod_{i=1}^{4} (\pm 1)_i = +1 \right\} \quad \text{(even number of minus signs)}, \tag{2}$$

$$\mathbf{8}_c = \left\{ \left(\pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2}\right) \;\middle|\; \prod_{i=1}^{4} (\pm 1)_i = -1 \right\} \quad \text{(odd number of minus signs)}. \tag{3}$$

2 The Full Level-l Fock Space Generating Function

The multi-variable partition function tracking the string oscillator mass level $q = e^{2\pi i \tau}$ and the SO(8) Cartan charges $x = (x_1, x_2, x_3, x_4)$ is:

$$\mathcal{Z}(q, x) = \prod_{n=1}^{\infty} \frac{\prod_{w \in \mathbf{8}_s} (1 + q^n e^{w \cdot x})}{\prod_{w \in \mathbf{8}_v} (1 - q^n e^{w \cdot x})} = 1 + \sum_{l=1}^{\infty} q^l \mathcal{F}_l(x). \tag{4}$$

Expanding order-by-order in q, the level-l character polynomial is:

$$\mathcal{F}_l(x) = \sum_{w} c_w \, e^{w \cdot x}, \tag{5}$$

where $c_w \in \mathbb{Z}_{\geq 0}$ is the exact multiplicity of states with SO(8) weight w at mass level $\alpha' M^2 = l$.

3 The Universal Massive Supersymmetry Factor $\mathcal{M}_{\text{susy}}$

At the massless level $l = 0$, the open string ground state forms the 10D $\mathcal{N} = 1$ super-Maxwell multiplet:

$$\text{Ground state: } \mathbf{8}_v \oplus \mathbf{8}_s \quad (8 \text{ bosons} + 8 \text{ fermions}). \tag{6}$$

For any massive level $l \geq 1$, the fermionic zero modes satisfy the Clifford algebra:

$$\{S_0^a, S_0^b\} = 2\,\delta^{ab} p^+. \tag{7}$$

Quantization of these zero modes gives the universal 256-state massive supermultiplet $\mathcal{M}_{\text{susy}}$, which corresponds to the tensor product:

$$\mathcal{M}_{\text{susy}} = (\mathbf{8}_v \oplus \mathbf{8}_s) \otimes (\mathbf{8}_v \oplus \mathbf{8}_s). \tag{8}$$

Decomposing into SO(8) irreducible representations:

$$\text{Bosons:} \quad (\mathbf{8}_v \otimes \mathbf{8}_v) \oplus (\mathbf{8}_s \otimes \mathbf{8}_s) = (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_v) \oplus (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_s) = \mathbf{44} \oplus \mathbf{84} = \mathbf{128}_B, \tag{9}$$

$$\text{Fermions:} \quad 2 \times (\mathbf{8}_v \otimes \mathbf{8}_s) = 2 \times (\mathbf{8}_c \oplus \mathbf{56}_c) = \mathbf{128}_F. \tag{10}$$

Therefore:

$$\mathcal{M}_{\text{susy}} = \mathbf{128}_B \oplus \mathbf{128}_F. \tag{11}$$

Every massive open string level $l \geq 1$ must factorize as:

$$\mathcal{F}_l = \mathcal{M}_{\text{susy}} \otimes r^{(l)} = (128_B \oplus 128_F) \otimes (r_B^{(l)} \oplus r_F^{(l)}), \tag{12}$$

where $r^{(l)}$ is the reduced "prefactor" representation. The physical state count is:

$$N_B = N_F = 128 \times \left(\dim r_B^{(l)} + \dim r_F^{(l)}\right), \tag{13}$$

ensuring exact spacetime supersymmetry at every massive level.

4 Exact Character Polynomial Division Algorithm

In terms of character polynomials, factoring out $\mathcal{M}_{\text{susy}} = (\mathbf{8}_v \oplus \mathbf{8}_s)^2$ is accomplished by dividing $\mathcal{F}_l(x)$ by the single linear character:

$$\chi_{\text{susy}}(x) = \chi_{\mathbf{8}_v}(x) + \chi_{\mathbf{8}_s}(x) = \sum_{w \in \mathbf{8}_v} e^{w \cdot x} + \sum_{w \in \mathbf{8}_s} e^{w \cdot x}. \tag{14}$$

The reduced character polynomial is:

$$Q_l(x) = \frac{\mathcal{F}_l(x)}{\chi_{\text{susy}}(x)}. \tag{15}$$

4.1 Division in Lexicographic Weight Order

The division is executed iteratively:

1. Let the remainder polynomial initially be $\text{rem}(x) = \mathcal{F}_l(x)$.
2. Find the highest active weight $W = (W_1, W_2, W_3, W_4)$ in $\text{rem}(x)$ in standard lexicographic order, with non-zero coefficient c_W.
3. The leading weight of $\chi_{\text{susy}}(x)$ is the highest weight of $\mathbf{8}_v$, which is $w_{\text{lead}} = (1, 0, 0, 0)$.

4. The corresponding quotient term is:

$$q_W(x) = c_W\, e^{(W - w_{\text{lead}})\cdot x}. \tag{16}$$

5. Update the quotient and remainder:

$$Q_l(x) \leftarrow Q_l(x) + q_W(x), \qquad \text{rem}(x) \leftarrow \text{rem}(x) - q_W(x)\chi_{\text{susy}}(x). \tag{17}$$

6. Repeat until $\text{rem}(x) \equiv 0$.

By spacetime supersymmetry, the division is exact and the remainder vanishes identically.

5 Decomposition into SO(8), SO(9), and SO(10) Representations

5.1 SO(8) Weyl Character Inversion

Using the Weyl denominator $\Delta_{D_4}(x) = \sum_{w\in\mathcal{W}_{D_4}} \det(w) e^{w(\rho)}$ with $\rho = (3,2,1,0)$, we invert the character formula:

$$Q_l(x)\cdot\Delta_{D_4}(x) = \sum_{\lambda} m_\lambda \left(\sum_{w\in\mathcal{W}_{D_4}} \det(w) e^{w(\lambda+\rho)} \right), \tag{18}$$

yielding the exact SO(8) representation breakdown:

$$r^{(l)} = \bigoplus_{\lambda} m_\lambda\, \mathbf{R}_\lambda^{\text{SO}(8)}. \tag{19}$$

5.2 SO(8) → SO(9) Massive Little Group Lift

Massive states in 10 dimensions must assemble into representations of SO(9). Using the Gelfand-Tsetlin branching rule:

$$(h_1,h_2,h_3,h_4)_{\text{SO}(9)} \longrightarrow \bigoplus_{g_1=h_2}^{h_1} \bigoplus_{g_2=h_3}^{h_2} \bigoplus_{g_3=h_4}^{h_3} \bigoplus_{g_4=-h_4}^{h_4} (g_1,g_2,g_3,g_4)_{\text{SO}(8)}, \tag{20}$$

we iteratively lift the SO(8) spectrum into SO(9) irreducible representations $r^{(l)}_{\text{SO}(9)}$.

5.3 SO(9) → SO(10) $D = 11$ Supermultiplet Assembly

Under Proposal III.B, the cumulative spectrum $R^{(n)} = \sum_{l=1}^{n} r^{(l)}$ lifts to representations of the $D = 11$ massive little group SO(10). The branching $D_5 \to B_4$ is given by:

$$(H_1,H_2,H_3,H_4,H_5)_{\text{SO}(10)} \longrightarrow \bigoplus_{h_1=H_2}^{H_1} \bigoplus_{h_2=H_3}^{H_2} \bigoplus_{h_3=H_4}^{H_3} \bigoplus_{h_4=|H_5|}^{H_4} (h_1,h_2,h_3,h_4)_{\text{SO}(9)}. \tag{21}$$

Crucially, when $H_5 = \pm 1$ (the chiral self-dual 5-form $(1,1,1,1,1) = \mathbf{126}$), h_4 is restricted to $h_4 = 1$, branching solely to $\mathbf{126}_{\text{SO}(9)}$ without generating lower-rank forms. This ensures non-negative multiplicities across all levels $n = 1,\ldots,10$.

6 Summary of Results (Levels 1 to 10)

Level n	$\sum_{l=1}^{n} r^{(l)}$ Status	SO(10) Bosons	SO(10) Fermions	Negative Mults	Solve Time
1	**PASSED**	1	0	0	$< 0.001\,\mathrm{s}$
2	**PASSED**	1	0	0	$< 0.001\,\mathrm{s}$
3	**PASSED**	1	1	0	$< 0.001\,\mathrm{s}$
4	**PASSED**	2	1	0	$< 0.001\,\mathrm{s}$
5	**PASSED**	4	2	0	$< 0.001\,\mathrm{s}$
6	**PASSED**	6	5	0	$< 0.001\,\mathrm{s}$
7	**PASSED**	11	7	0	$0.001\,\mathrm{s}$
8	**PASSED**	19	10	0	$0.001\,\mathrm{s}$
9	**PASSED**	25	14	0	$0.003\,\mathrm{s}$
10	**PASSED**	34	19	0	$0.005\,\mathrm{s}$

Table 1: Exact verification of Proposal III.B for Levels 1 through 10.